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TryHackMe Writeup

Date Completed: June 1, 2026

Room Title: John the Ripper: The Basics

Difficulty: Easy

Link: <https://tryhackme.com/room/johntheripperbasics>

Focus Area: Hash Cracking

Executive Summary

Completed the **John the Ripper: The Basics** room on TryHackMe, gaining hands-on experience with one of the most powerful and widely used password cracking tools in cybersecurity. The room covered hash identification, cracking various hash formats (including NT hashes, Linux shadow files, and raw MD5), and extracting/cracking passwords from common encrypted file types such as ZIP, RAR, and SSH private keys (id_rsa).

By leveraging wordlists, single-crack mode, and custom rules, I was able to recover multiple weak passwords, escalate privileges, and access protected resources. This exercise significantly strengthened my practical skills in password auditing, hash cracking methodologies, and understanding common password weaknesses.

Learning Objective

- Explore the different hash cracking uses that John offers
 - Crack the hashes for common file types
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Tools & Technologies Used

Category	Tools/Technologies
Hash Cracking	John the Ripper

Walkthrough

Task 1-3: Introduction and information sections.

Task 4: Utilize the hash-id.py program to identify the hash types of different hashes and then use John to determine the password by running it through the rockyou.txt list.

Ex: john -format={hash type} -wordlist={path to wordlist} {path to hash}

Task 5: Cracking the NTHash that Windows uses. Use the same john command but change the hash type.

Ex: john -format=nt -wordlist={path to wordlist} {path to hash}

Task 6: Cracking Linux's /etc/shadow by extracting the hashes from the file. Use the unshadow command and then crack using the john command

Ex: unshadow local_passwd local_shadow > unshadowed.txt

John -format={hash} -wordlist={wordlist} unshadowed.txt

Task 7: People may use their username to decide their password. John can take the username and hash and run a single crack mode to rearrange the username into different variations to try and determine the password. The hash file needs to have the following format: username:hash

Ex: john -single -format=raw-md5 hash.txt

Task 8: This task teaches the custom rule formatting and asks various questions related to syntax. Humans are rather predictable and we can establish password patterns such as capitalizing the first letter in a password and ending our passwords with numbers and/or special characters. This predictability takes some of the complexity out of cracking passwords.

You can call a rule when using john with the flag "-rule={Rule Name}"

Task 9: John can be used to crack the hash of a password-protected ZIP file. To do this, use the zip2john command to save the password to a text file and then use john to crack it.

Ex: zip2john file.zip > hash.txt

John -wordlist={wordlist} hash.txt

Task 10: John can also crack the hash for RAR archives. Using the rar2John tool.

Ex: rar2john file.zip > hash.txt

John -wordlist={wordlist} hash.txt

Task 11: John can be used to crack the SSH keys held in the id_rsa files. Id_rsa utilizes a private key to bypass password authentication. A password is used to access the id_rsa file directly. You can use ssh2john.py, which is a Python program.

/opt/john/Ssh2john.py file > hash.txt

Task 12: Conclusion

Challenges Faced & Solutions

Challenge	Solution / Lesson Learned
Unzip zip file via cmd line	Used google to determine the right command to unzip via command line
Unzip rar file via cmd line	Used google to determine the right command to unrar via command line

Screenshots showing room completion

The screenshot shows a user profile for 'remin005' with a rank of 163127 (top 7%) and 10 badges. The user has completed 66 rooms and has a streak of 1. The profile includes buttons for 'Add Socials', 'Add Calendly Link', and 'Share room badges'. Below the profile, a navigation bar shows 'Completed rooms' as the active tab, with other options like 'Certificates', 'Skills matrix', 'Badges', 'Created rooms', 'Yearly activity', and '2025 ReCapMe'. The main content area displays two room cards: 'John the Ripper: The Basics' (Premium, Easy) and 'Metasploit: Introduction' (Free, Easy). A pagination bar at the bottom shows 'Previous', '1', '2', '3', '4', '5' (selected), and 'Next', with a note '5 of 5 available pages'.

The screenshot shows the 'John the Ripper: The Basics' room completion page. The room is marked as 'Premium room' and 'Room completed (100%)'. The page lists 12 tasks, all of which are completed (indicated by green checkmarks):

- Task 1: Introduction
- Task 2: Basic Terms
- Task 3: Setting Up Your System
- Task 4: Cracking Basic Hashes
- Task 5: Cracking Windows Authentication Hashes
- Task 6: Cracking /etc/shadow Hashes
- Task 7: Single Crack Mode
- Task 8: Custom Rules
- Task 9: Cracking Password Protected Zip Files
- Task 10: Cracking Password-Protected RAR Archives
- Task 11: Cracking SSH Keys with John
- Task 12: Further Reading

At the bottom, there is a recommendation section asking 'How likely are you to recommend this room to others?' with a scale from 1 to 10.